

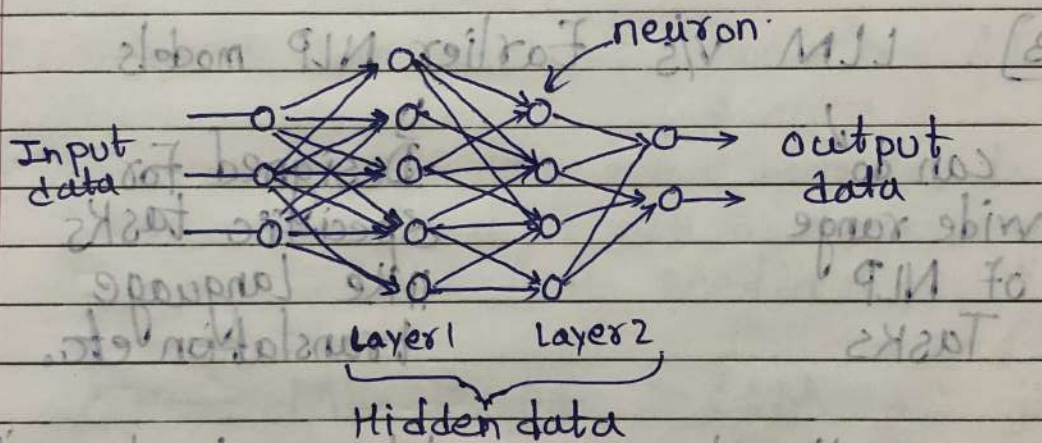
Intro to LLM

1] LLM

↳ is Deep Neural Network,

Trained on massive amount of Data
to Understand, generate and respond to
Human like text

• Deep neural Network trained on Massive
amount of text data for LLM



2] Large Language Model (LLM)

Models
have billions
of Parameter

These model Only deals with language
do a wide range of NLP tasks?
question-answering, translation,
sentiment analysis and much
More!

Ex. →

GPT-3 small → 125 Million parameter

GPT-3 175B → 175 Billion parameter

The latest No. of parameter
is close approaching trillion

As LLM models Upgrading

↳ No. of parameters
No. of Tokens
No. of Decoders } Increasing significantly

In 1980 → NLP have 1-10 parameters

In 2020 → it reached almost 1 trillion parameters

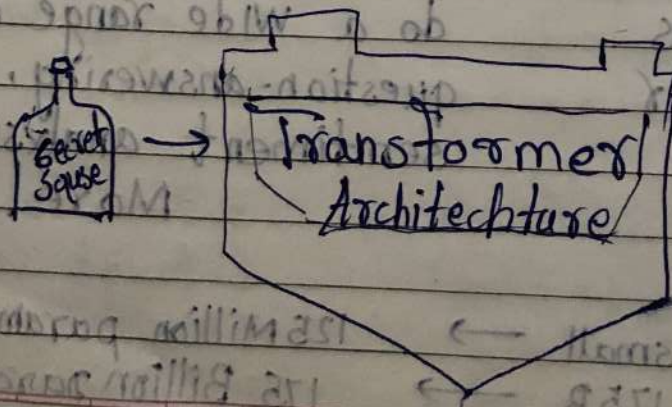
3) LLM v/s Earlier NLP models

can do
wide range
of NLP
Tasks

Designed for
Specific tasks
like language
translation etc.

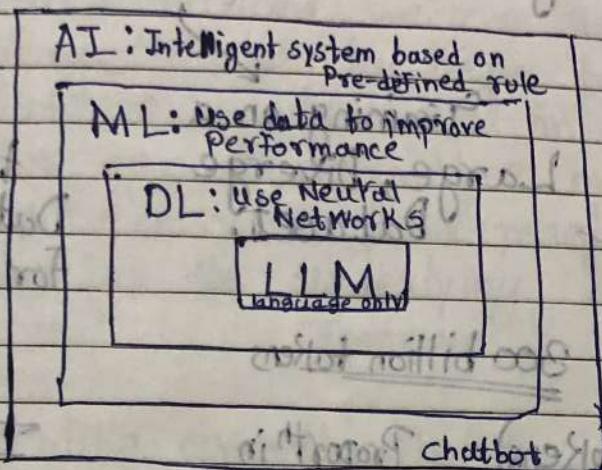
Eg. Earlier language model could not write
an email from custom instructions,
a task that is trivial for Modern LLMs

4) What makes LLMs so good?



we will see
further

5] LLM V/s Generative AI V/s Deep Learning V/s Machine Learning
 V/s AI



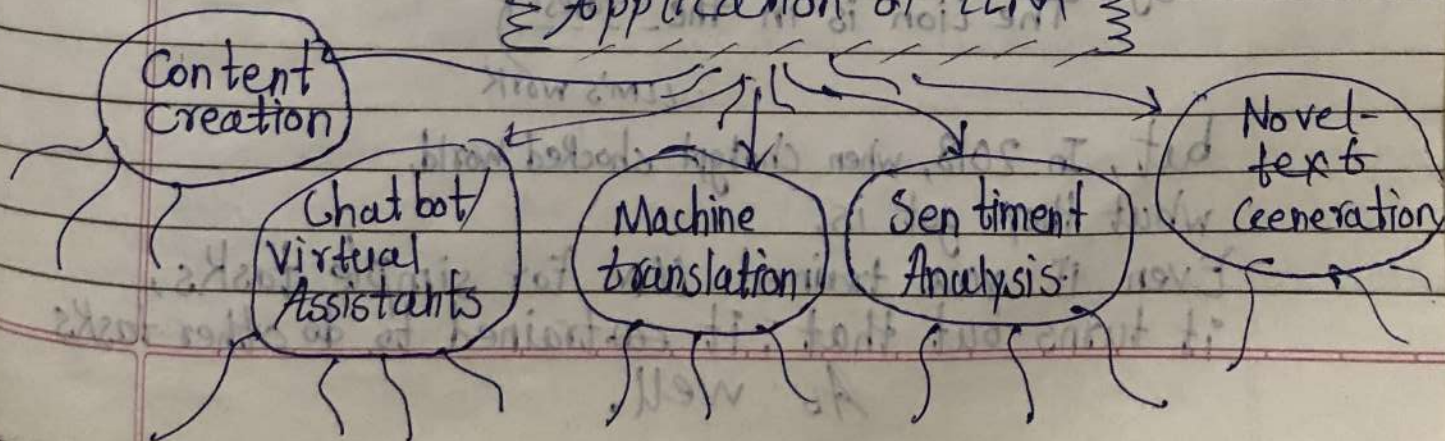
LLM + Deep learning : Generative AI

use deep neural networks to create
 New content such as text, images, Various
 form of Media

AI > ML > DL > LLM

∴ LLMs represent a specific application of deep
 learning techniques, leveraging their ability
 to process & generate human like text

Application of LLM



Stages of Building LLM

Creating an LLM = Pretraining + Finetuning
(Foundation model)

Training on a Large, Diverse Dataset

Refinement by training on narrow Databases, Specific for particular task

Chatgpt-3
↓
trained on 300 billion tokens

Dataset	Quantity (tokens)	Proportion in training data
---------	-------------------	-----------------------------

Common crawl	410 billion	60%
webtext 2	19 billion	22%
Book1	12 billion	8%
Book2	35 billion	8%
Wikipedia	3 billion	3%

It is necessary
If you want answer
pre or any output
on specific or costume
task or domain

↓
If want to deploy
LLM operations
application on your
Individuals dataset

You need
Finetuning

Initially: LLM are trained to predict
the next word in sentence

eg. The Lion is in the forest

LLM's work

but, In 2018, when chatgpt shocked world,
what they got is,
Even if you trained LLM for simple tasks,
it turns out that; it ~~can~~ trained to do other tasks
As well.

Blog-post From Open-AI

G. We also noticed we can use the underlying language model to begin to perform tasks without ever training on them.

For eg. Performance on tasks like picking the right answers to a question steadily increases as the underlying language model improves.

So, it like even though you trained LLM for predicting the next word, it turned out that it can do variety of other things like translation, answering-questioning mcq, Sentiment Analysis, winograd schema Resolution, linguistic Acceptability, etc

It is possible, because it trained on Huge No. of data

Only pretraining is sufficient, for someone who just want basic to get stuff present on Network, like student

Finetuning is for costumizable conditions

For ex. lawyer : Who want study of only legal case studies,

So while Foundatⁿ models were strong at reasoning, they lacked the extensive knowledge of legal case history & other knowledge required for legal work

Eg. 7A-1990 most top-fold

1) Airline Industry:

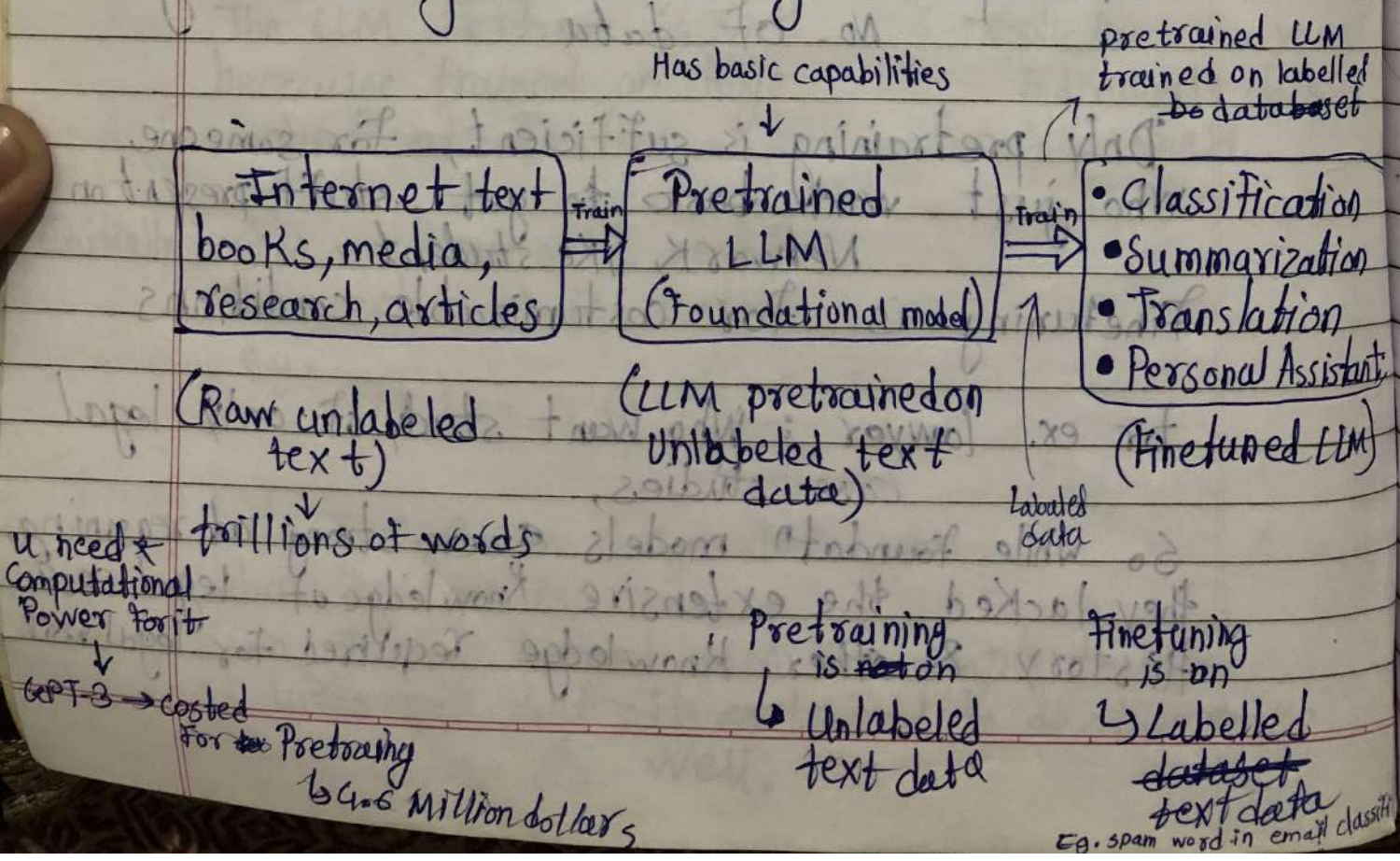
SK Telecom Fine-Tuned GPT-4 to improve customer service interactions, specially for telecom related conversation in Korean.

News: 2) JP Morgan Chase Unveils AI powered LLM suite;

↳ so their it's GPT-4 already then why their own
↳ cause they Fine-tuned from their own data, for their employee specifically

cause The JP Morgan's data is not publically available only they have it, So the answers are specific for their company

Pretraining + Finetuning Schematic



Rebooting is not possible Unless
You have access to GPU & 4.6 Million \$

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Steps For Building a LLM

1] Train on a large corpus of text data

(Raw text = Regular Text without any labelling info.)

2] First training stage of LLM is also called Pretraining

↳ creating an initial pretrained LLM (Foundation model)

EX. GPT-3 model is a pretrained model which is capable of text completion.

But it also turned out, it is able to do much more.
So

Pretraining on large data gives more power to LLM

3] After obtaining the pretrained LLM, we can further train LLM on labelled data (Finetuning)

4] There are 2 categories in Finetuning

Instruction
Finetuning



Labeled dataset

consists of instruction-answer pairs.

eg. text-translation,
airline customer support

Finetuning for
classification tasks



Labeled dataset

consist of text
i associated labels.

eg email → spam v/s no
spam

lecture - 89

Basic Intro to Transformers

1) Most Modern LLMs rely on the Transformer Architecture.

Transformer Architecture } Deep Neural Network Architecture
introduced in 2017 paper
(Attention All you need)

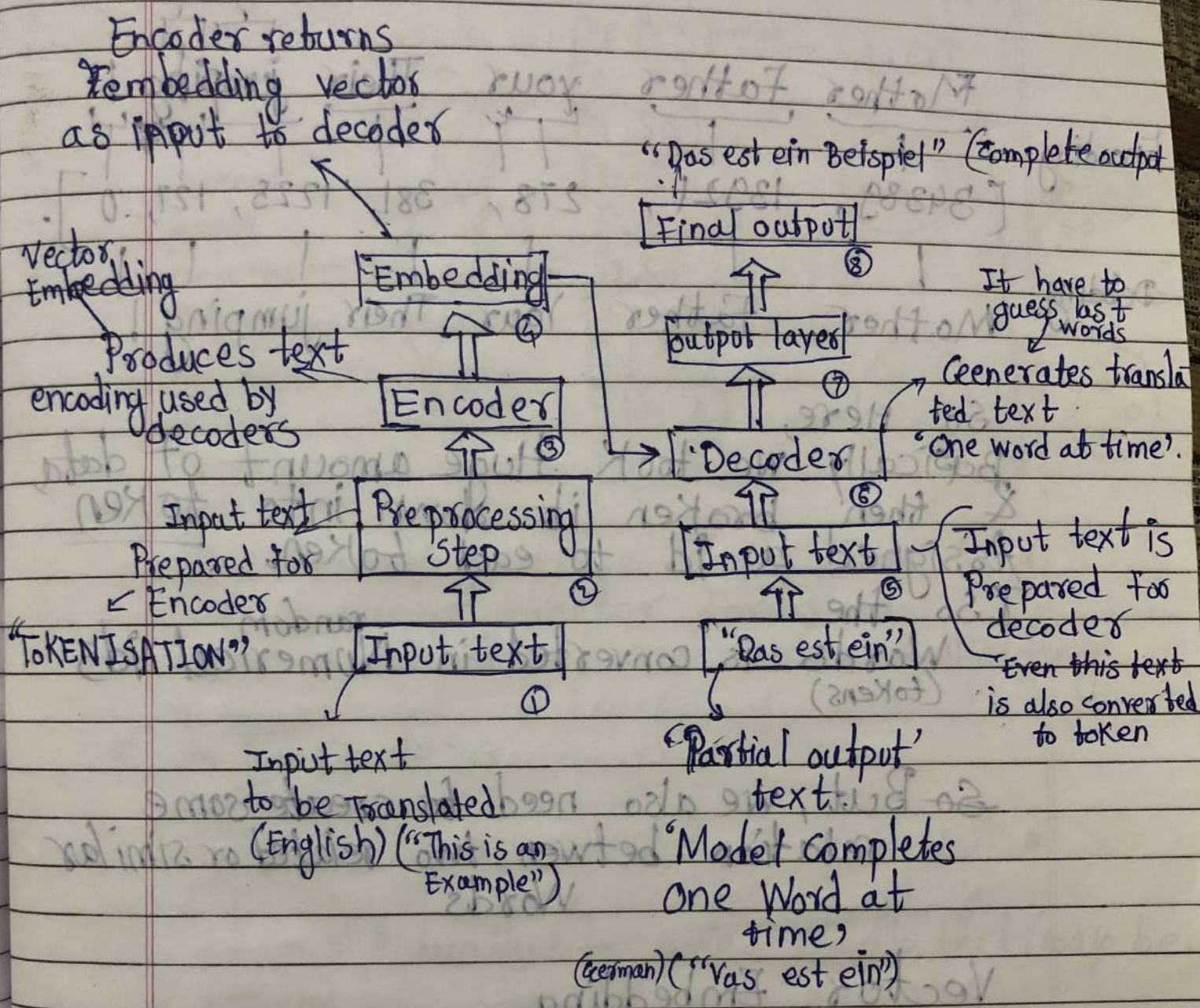
Attention All you need

When this architecture ~~is~~ was building
It was actually proposed for Translation.
task, the chatgpt Architecture is nothing in idea.

Later it is derived that, this Transformer Architecture
can do a lot of stuff

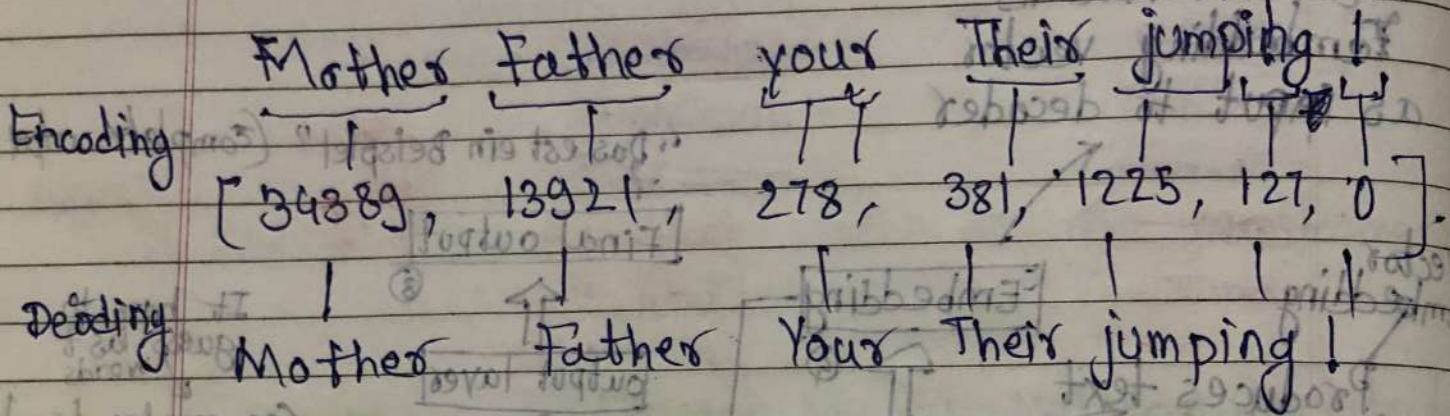
SO, Original transformer is Developed for
English → German/French
Translation

Simplified Transformer Architecture



Actual Transformer Not look like this,

Tokenize the Data.



So Here,
Basically we took Huge amount of data
& then broken it down into tokens
Assigned a id to each token.
So the Words is converted in ^{random} Numerical ids
(tokens)

So But, we also need to create some
relation between two related or similar
Words

Vector Embedding

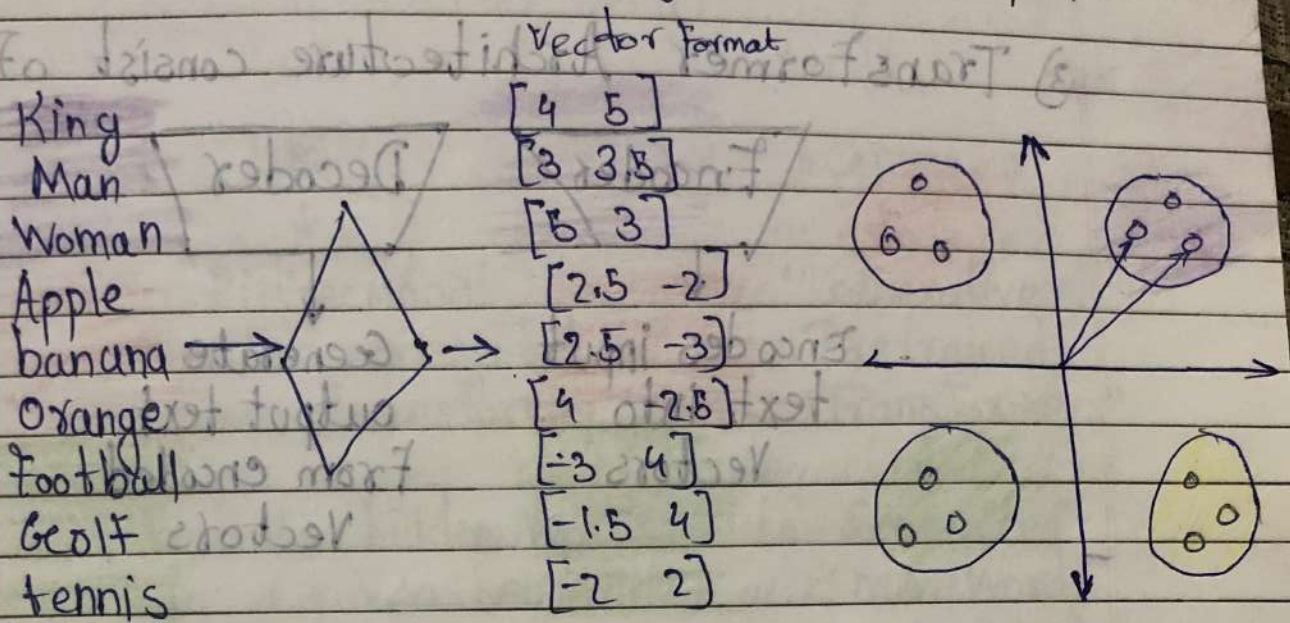
Representing tokens in a way, which
Captures the semantic meaning in the
word.

This process is Known as Vector Embedding

Example!

Range (128 to 4096)
 It can go in few thousand dimensional space

Vector Embeddings (2D Example)



So converting these words into this Vector Forms

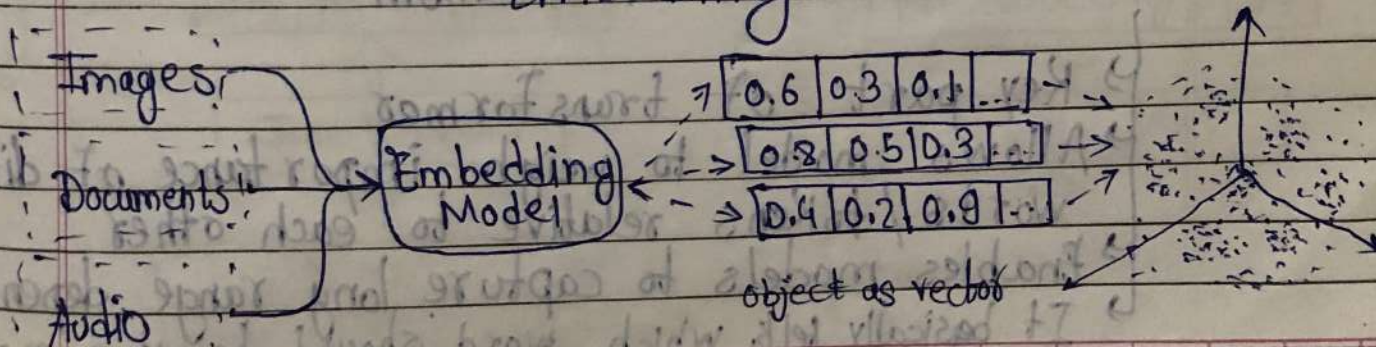
is known as vector Embedding

* so Encoders

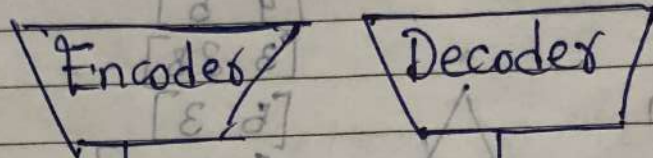
↳ Main purpose is

↳ Actually to take Input text (may be tokens)

& convert those tokens into Vector Embedding



3) Transformer Architecture consist of



Encodes input text into Vectors

Generate output text From encoded Vectors

- Attention Mechanism: Allows you to model dependencies without regard to their between Different words, without regards to How close apart or far apart the words are.

- Self-attention: It is an Attention mechanism relating different position of a single sequence in order to compute a representation of the sequence.

Self-Attention mechanism

- ↳ Key part of transformer
- ↳ Allows model to weigh importance of different words / tokens relative to each other
- ↳ Enables models to capture long range dependencies
- ↳ It basically tells which word should be given more attention, when you predict the next word

4) Later variations of transformer Architecture (After 2016)

BERT

↓
"Bidirectional encoder representation from transformers"

↓
Predicts hidden words in a given sentence

↓
Predicts hidden words in a given sentence

GPT

↓
"Generative Pretrained Transformers"

↓
Generates new words

↓
Generates New words

BERT

GPT

Fill Missing words

Leads to generate one word at a time

This is an example of how LLM can perform

This is an example of how LLM can perform

Encoder

Decoder

Preprocessing Steps

Pre Processing steps

Input text

Input text

This is an - - - of how LLM - - - perform

This is an example of how LLM can - - -

Receives input where words are randomly masked during training

Receives Incomplete text

in BERT,
Random words can be
Masked

In GPT,
Right most part of
Sentence is predicted

∴ BERT Actually do
very well, in
Sentimental analysis

GPT read
left to Right to predict
Right side things.

∵ Cause, the
Bidirectional Property
By looking at the entire
Sentence from both directions
BERT can capture nuances &
Relationship between words that are
Important for understanding meaning
& context.

For ex. it can different between 'bank' as
riverbank & Financial institution

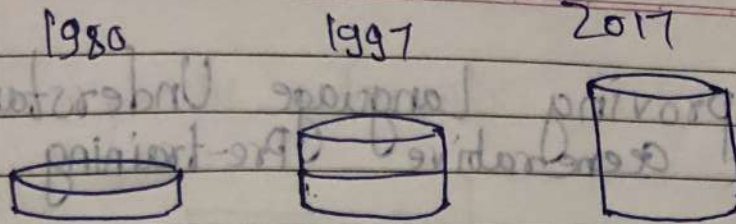
5) Transformer V/s LLM

↳ Not all transformers are LLMs

↳ Transformers can also be used for
Computer Vision

↳ Not all LLMs are transformers

↳ LLM can be based on recurrent or
Convolutional architectures as well



Recurrent
Neural
Network

Long-short
term
Memory
Network

Transformers

LLM
can be
Recurrent
Neural Network
or Long-short term
Memory Network

LLMs can be based on recurrent or convolutional architectures as well.

Not all LLM's are Transformers.

Transformers & supervised pre-training

Labels are not given to training data.

The labels were taken from the sentences itself. We do not need to pre-label the data.

So at this point, researchers are interested in this paper. But a lot of work is still to be done.

Second
Paper
by open AI

Improving Language Understanding by Generative Pre-training

They took the Transformer^{Architecture}, they removed the encoder block, & trained the model on huge amount of dataset to just predict the next words in an unsupervised manner.

It could really improve language understanding

they called the training phase as generative pre-training

↳ why generative

↳ cause we generating next word

So, openAI approach is

↳ Transformers & unsupervised pre training

↳ Means

labels are not given to training data.

↓

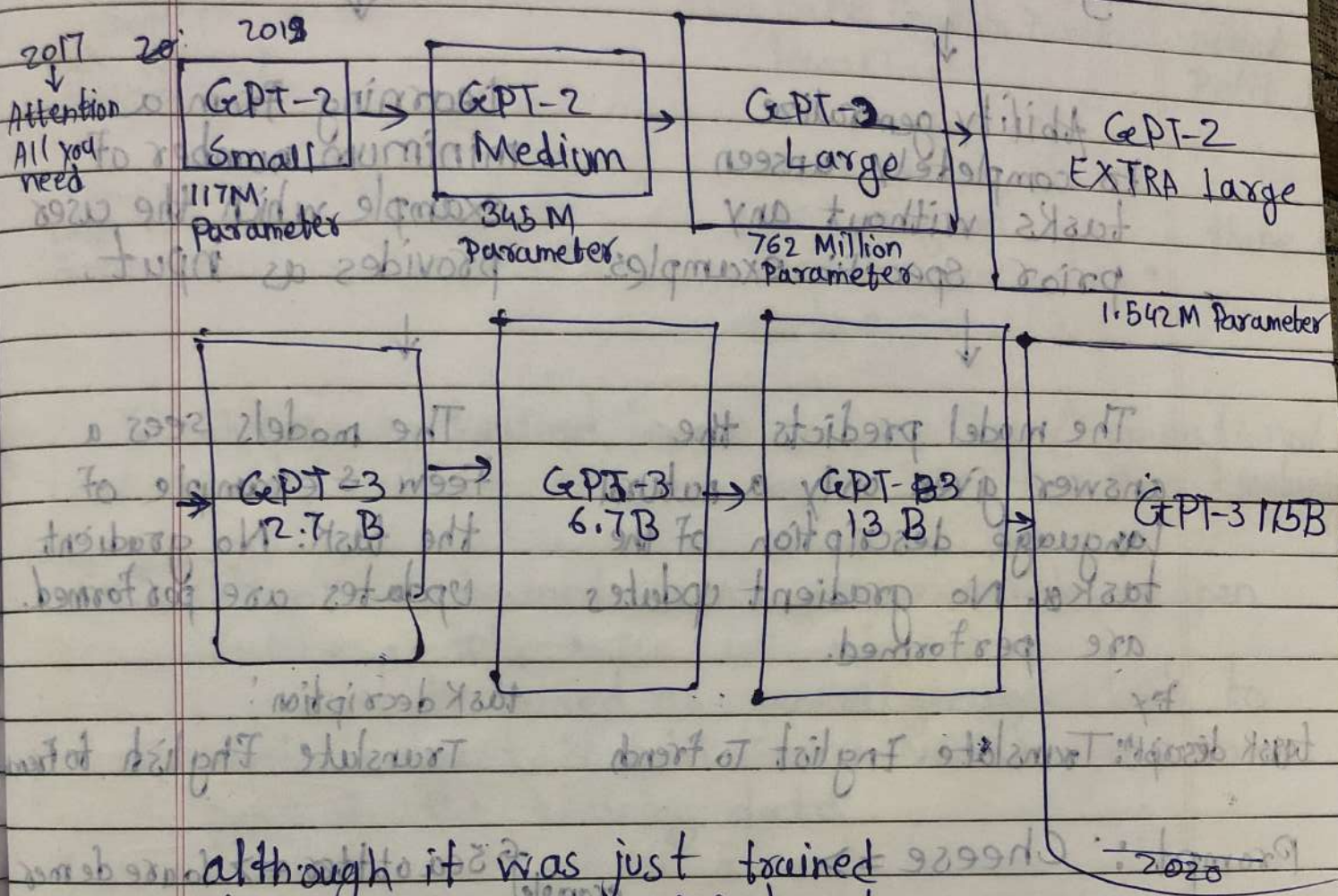
The labels were taken from the sentences itself we do not need to pre-label the data.

So at this point,

researchers are interested in this paper

But on commercial No one is observing it

Then Open AI released



although it was just trained to predict next word it can do number of other things like translation sentiment analysis, answering questions, answering mcq, emotional recognition, many things.

Transformers → GPT → GPT2 → GPT3 → GPT 3.5 → GPT4 (2023)
 (2017) (2018) (2019) (2020) (2022)
 GPT-4 Turbo (2023)

Transformer V/S GPT

GPT essentially borrows from the Transformer architecture but it's a bit different in that it does not have really the encoder block

Zero SHOT v/s Few SHOT learning.

Ability generalize to completely unseen tasks without any prior, specific examples.

learning from a minimum number of example which the user provides as input.

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

The model sees a few examples of the task. No gradient updates are performed.

task description: Translate English to French

task description: Translate English to French

Prompt: Cheese $\Rightarrow$ ~~from~~ ^{example} Sea otter $\Rightarrow$ la arse demer
peppermint $\Rightarrow$ menthe poivrée
plush giraffe $\Rightarrow$ girafe peluche

Prompt: cheese $\Rightarrow$

GPT-3 was few shot learner

GPT-4 ~~can~~ can do few shot learning
by creating more suitable ^{better} prediction
but it also
can do a zero shot learning

Ex

Zero shot

Complete task without Example

Translate English to French
"breakfast"

output
Petit

Few shot

Complete task with Example

goat → goat
shoe → shoe
phone → phone

phone

* These pretrained models are base / Foundational model which can be used for further fine-tuning

* Many pretrained LLM's are available as open source models → can be used as general purpose tools to write, extract & edit text which was not part of the training data

Open Source GPT

v/s

Close source GPT

refers to GPT model whose source code, model, weights & sometimes training data are publically available. Anyone can access, modify, & use the model, usually under an open source licence.

Eg. GPT-2

refers to GPT models where, the code, model weights & sometimes training data are proprietary & not available to public. Access to model controlled & users typically interact with it via an API or specific platform eg. (websites)

Eg. GPT 4, GPT 3

Control safety, support & integration

Advantage: Customizable, cost

* GPT Architecture ← already drawn BERT v/s GPT

↳ GPT 3 is a scaled-up version of the previous model, implemented on a larger database

- GPT models are simple trained on 'next-word' prediction tasks

The lion roams in the jungle

next word

Example: Input (Features) Correct output (Labels)

1 Second law of robotics: a

↓

2 second law of robotics: a robot

↓

3 Second law of robotics: a robot must

• Output of previous prediction, is part of Next input

∴ The sentences broken down into input & output

The Input can 3-4-5 words but Output is only 1 word

That why it called as Unsupervised Learning

↳ because, we do not give labels the sentence itself consist of the labels, because the label is the next one

& because it requires huge amount of data, that data or sentences need to be broken down into tokens & labels, which require huge amount of compute power

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With this training, they can do a wide range of other tasks like translation, spellings correction, etc.

how is it trained

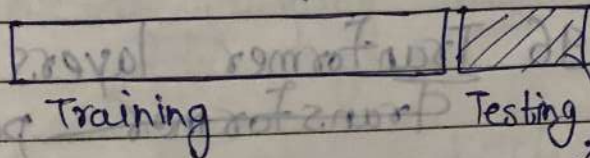
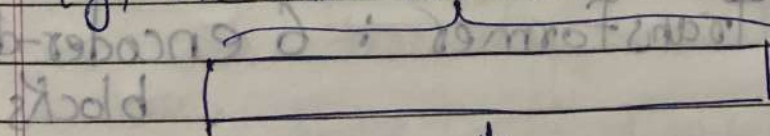
↳ So basically initially it predicts wrong outputs but then we have the correct output, which we know what should be the next word from the sentence itself,

So then the error will be computed based on the predicted output & the difference between the corrected output & then similar to the Back propagation done in neural network the weights of the transformer of the GPT Architecture will adapt so that the next word is predicted correctly

• Next word prediction: Self Supervised Learning

self labeling

Eg, We have sentence



... that's why it call unsupervised learning, cause no need to label externally it is labeling itself

by using this input it will try to predict the next word

next true word (which we know)

Predicted word

GPT Architecture Trained to Minimize difference between these two & Update the weights

∴ We don't collect labels for training data, but use the structure of data itself.

↳ Next word in sentence is used as label

↳ That's why it called Autoregressive Model

↳ Use previous output as input for future prediction

∴ Pre training part of GPT is ^{consist} of
↳ Unsupervised
↳ Autoregressive

• GPT Architecture have no encoder, we just have the decoder.

It is simpler than Transformer

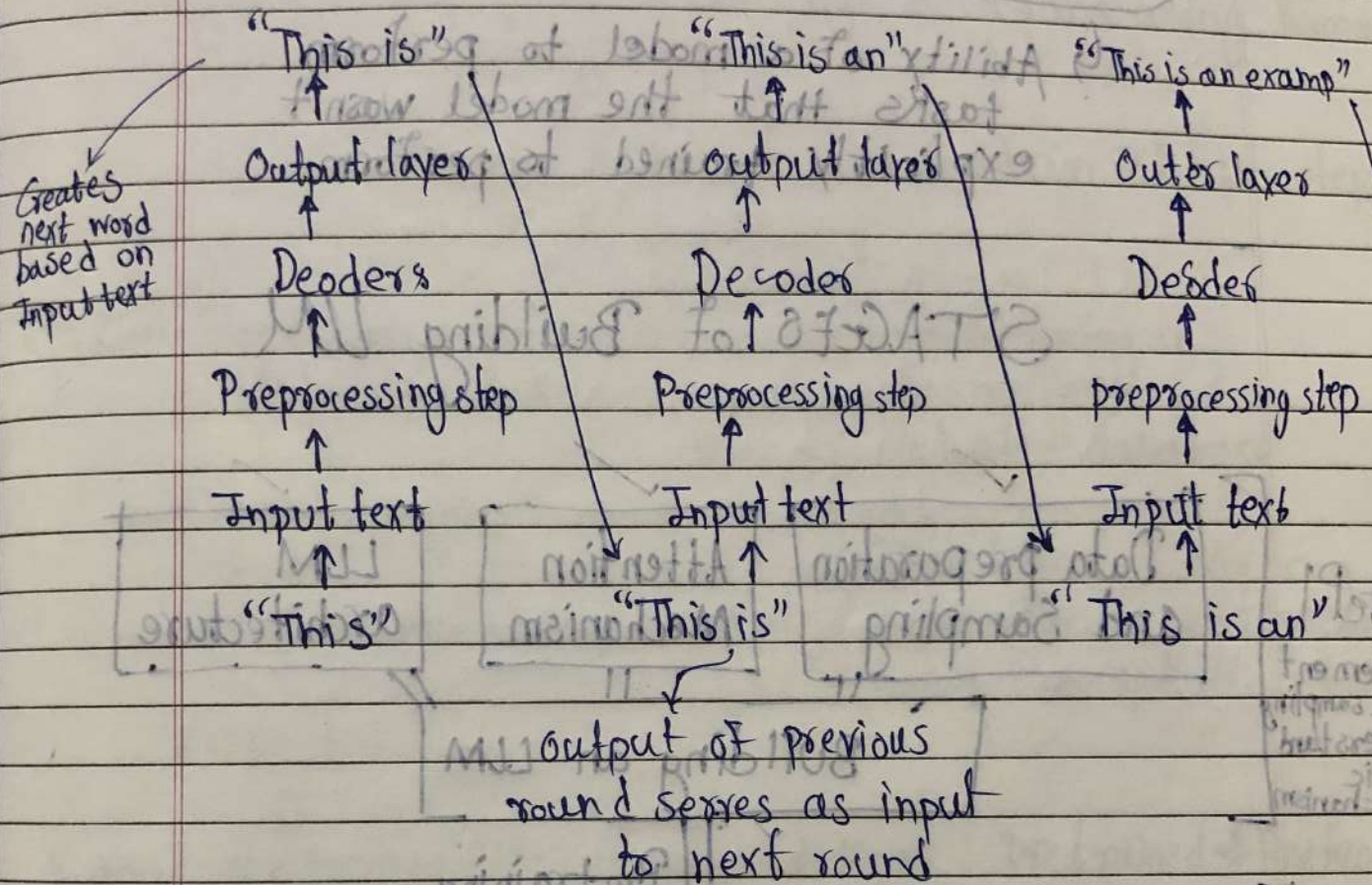
• Digital Transformer : 6 encoder-decoders blocks

↓
• GPT 3 : 96 Transformer layers, & 175 Billion Transformer parameters

Iteration 1

Iteration 2

Iteration 3



Although GPT only trained, Only for next word prediction
 so may other task
 like translation & much more
 cause,
 while the training happens, it develop other
 Advantages.

It develop other capabilities. This property is called
Emergent Behaviour.